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AUTOMATED OPTICAL STRUCTURAL HEALTH MONITORING (SHM) SYSTEM

REAL-TIME AIRCRAFT SURFACE DAMAGE TOLERANCE AND DEFECT EVALUATION WEB APPLICATION USING OPENCV, NUMPY, AND STREAMLIT

A project report submitted in partial fulfillment of the requirements for the course

AERO-STRUCTURAL HEALTH MONITORING & NON-DESTRUCTIVE TESTING
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DECLARATION

I hereby declare that the project report entitled **"Automated Optical Structural Health Monitoring (SHM) System: Real-Time Aircraft Surface Damage Tolerance and Defect Evaluation Web Application using OpenCV, NumPy, and Streamlit"** is based on the project work carried out by me as part of the academic curriculum of Jawaharlal College of Engineering and Technology.

The report has been prepared for academic purposes. External materials, concepts, fracture mechanics formulations (Paris' Law, LEFM), and references used in preparing this report have been acknowledged appropriately. I understand that the report is subject to institutional academic integrity requirements.

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ABSTRACT

Modern operating aircraft structures sustain continuous cyclic flight loading and environmental degradation over lifespans exceeding 25 to 30 years. Primary load-bearing airframe panels—specifically 2024-T3 aluminum fuselage skins, 7075-T6 wing skins, and Carbon Fiber Reinforced Polymers (CFRP)—are prone to fatigue micro-cracking at fastener stress risers ($K_t \approx 3.0$), exfoliation corrosion pitting, and composite delamination. Conventional manual visual inspection (VT) during heavy maintenance checks is labor-intensive, subjective, and prone to human inspection fatigue.

The **Aero-SHM System** addresses this airworthiness challenge by coupling a modular 9-stage Computer Vision (CV) pipeline with an interactive Python and Streamlit web application. Operating locally with minimal dependencies, the application processes high-resolution optical surface scans using Contrast-Limited Adaptive Histogram Equalization (CLAHE) for hairline crack shadow amplification, 2D Gaussian spatial smoothing, Canny hysteresis edge tracking, and adaptive thresholding for localized pitting corrosion. An automated Hough Circle suppression module eliminates circular rivet heads to prevent false defect alarms.

For each detected anomaly, the engine extracts invariant geometric descriptors including bounding box coordinates [X, Y, W, H], surface area, aspect ratio, and solidity. It computes a global Defect Pixel Density percentage (D%) and assigns dynamic airworthiness status badges aligned with **FAA Advisory Circular AC 25.571-1D** (Healthy, Caution, Critical). Additionally, it integrates Linear Elastic Fracture Mechanics (LEFM) and **Paris' Law** ($da/dN = C(\Delta K)^m$) to predict remaining flight cycles to catastrophic brittle failure (N_f). A procedural synthetic specimen generator is incorporated for out-of-the-box laboratory testing. On automated benchmark suites, the pipeline executes in 77 ms per frame with 100% detection accuracy across test specimens.

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Chapter 1
INTRODUCTION

1.1 Background

Modern digital signal processing and computer vision algorithms have introduced major advancements in aerospace Non-Destructive Inspection (NDI). In commercial transport aviation, airframe structures operate under continuous cyclic flight pressurization loads, gust buffeting, and severe thermal and environmental cycles. High-strength aluminum alloys such as 2024-T3 Alclad and 7075-T6, along with advanced Carbon Fiber Reinforced Polymers (CFRP), constitute the primary load-bearing members. Ensuring the damage tolerance and airworthiness of these materials requires automated computational inspection tools capable of detecting sub-millimeter cracks and localized corrosion.

1.2 Introduction to the Proposed System

The Aero-SHM System is an interactive, full-stack computational platform engineered with Python, Streamlit, OpenCV, NumPy, and Matplotlib. It bridges digital image processing with aerospace structural mechanics. The system accepts high-resolution optical imagery, removes ambient lighting gradients, applies Contrast-Limited Adaptive Histogram Equalization (CLAHE) to boost hairline fracture contrast, filters rolling grain striations via Gaussian smoothing, and tracks boundaries using Canny hysteresis edge detection. Circular fastener rivets are filtered via Hough Circle transforms. Real-time engineering metrics—including defect pixel density, bounding box coordinates, aspect ratios, and Paris' law fatigue predictions—are immediately displayed on an interactive dashboard.

1.3 Problem Statement

The visual inspection of primary aircraft skin currently presents severe challenges:

- **Human Inspector Fatigue:** Manual visual inspection (VT) during depot-level C-checks requires examining thousands of square feet of riveted panels under varying lighting conditions.
- **Fastener Hole Stress Concentrations:** Microcracks initiate at fastener countersinks ($K_t \approx 3.0$) and remain undetectable to the naked eye until reaching several millimeters in length.
- **Absence of Quantitative Metrics:** Traditional visual inspection logs qualitative notes without recording precise pixel damage ratios or bounding box coordinates.
- **Decoupled Fracture Mechanics:** Existing inspection tools do not link real-time detected crack lengths directly with Paris' Law cyclic fatigue life predictions.

1.4 Reason for Deploying the System Through Streamlit

Deploying the computer vision engine through a Streamlit web application provides significant advantages: (1) *Lightweight, decoupled architecture* allowing direct in-memory array manipulation; (2) *Interactive hyperparameter tuning* for real-time adjustments of CLAHE and Canny thresholds; (3) *Local execution and data privacy* ensuring sensitive aircraft drawings and photos remain on the local machine; and (4) *Cross-platform browser accessibility* at http://localhost:8501 without complex client installations.

1.5 Objectives

1. To construct an automated 9-stage digital image processing pipeline in OpenCV for surface defect detection.
2. To classify structural anomalies into Fatigue Cracks, Corrosion Patches, and Composite Scratches using geometric invariants.
3. To calculate real-time Defect Pixel Density (%) and assign FAA AC 25.571-1D airworthiness badges.
4. To implement an interactive Paris' Law LEFM calculator estimating remaining flight cycles to failure (Nf).
5. To develop a procedural synthetic specimen generator for offline, out-of-the-box laboratory testing.

Chapter 2
LITERATURE SURVEY

2.1 Limitations of Conventional Approaches

A critical review of aerospace Non-Destructive Testing (NDT) methodologies demonstrates severe operational trade-offs: Manual Visual Inspection (VT) accounts for over 80% of airline NDI tasks, yet detection reliability decreases sharply for crack lengths below 4 mm under specular aluminum glare. High-Frequency Eddy Current (HFEC) testing is sensitive to conductive fastener cracks, but requires slow, point-by-point probe alignment. Phased Array Ultrasonic Testing (PAUT) requires messy couplant gels and complex transducer calibration, making rapid large-area exterior screening impractical.

2.2 Comparative Analysis of NDT Methodologies

Dimension	Manual VT	HFEC	PAUT	Aero-SHM (Proposed)
Physical Mechanism	Human vision	Electromagnetic	Acoustic wave	Digital CV / Luminance
Inspection Speed	Moderate	Slow (10s/hole)	Slow	Real-Time (77 ms/panel)
Surface Contact	No	Yes (Probe)	Yes (Couplant)	No (Non-contact)
Measurement	Manual scale	Voltage signal	Time of flight	Sub-mm pixel pitch
Classification	Subjective	Crack only	Void only	Crack/Corrosion/Scratch
LEFM Link	None	None	None	Paris' Law (Nf)
Audit Log	Manual logbook	Screenshot	Scan file	CSV with [X,Y,W,H]

Table 2.1: *Comparative analysis of aerospace Non-Destructive Testing (NDT) methods*

Chapter 3
SYSTEM REQUIREMENTS

3.1 Hardware Requirements

- Standard desktop or laptop with dual-core 2.0 GHz+ processor (Intel Core / AMD Ryzen).
- Minimum 4 GB RAM for high-resolution image array buffering (8 GB recommended).
- 500 MB free storage for source code, virtual environment, and verification artifacts.

3.2 Software Requirements

- Operating System: Windows 10/11, macOS 12+, or Ubuntu Linux 20.04+.
- Python 3.10 / 3.11 with Streamlit (>=1.35.0), OpenCV (>=4.9.0), NumPy (>=2.0.0), Matplotlib, Pandas.

3.3 Development Environment

The application executes locally using Streamlit's native server at **http://localhost:8501**.

Chapter 4
SYSTEM DESIGN

4.1 System Architecture

The system architecture decouples user interface from signal processing. Input buffers (uploaded images or synthetic specimens) enter the SHM Computer Vision Pipeline, undergoing sequential filtering and morphological fusion. Segmented contours are classified by the aeromechanical classifier, while global metrics feed into the LEFM Damage Tolerance Engine. The Streamlit dashboard renders the final composite views.

4.4 Project Directory Structure

```
aero_shm_app/
|-- app.py                  (Streamlit web dashboard and interactive workbench)
|-- shm_cv.py               (Core 9-stage OpenCV computer vision pipeline)
|-- mock_generator.py       (Procedural aircraft specimen generator)
|-- utils.py                (Aerospace materials DB, Matplotlib plotting, CSV)
|-- verify_shm.py           (Automated test suite and validation engine)
|-- push_to_github.py       (Pure-Python Git remote sync utility)
|-- requirements.txt        (Minimal dependency specifications)
|-- README.md               (Engineering guide and presentation documentation)
`-- verification_artifacts/ (Validation test reports and annotated inspection images)
```

Listing 4.1: *Project directory structure*

Chapter 5
METHODOLOGY

5.1 Overview & Defect Classification Taxonomy

Table 5.1 defines the morphological criteria used to classify detected structural defects.

Defect Class	Morphological Criteria	Aeronautical Failure Mechanism
Critical Fatigue Crack	AR >= 3.0, Solidity < 0.65, Length >= 12mm	Cyclic tension hoop stress at fastener notch (Kt ≈ 3.0)
Developing Fatigue Crack	AR >= 3.0, Solidity < 0.65, 5 <= Length < 12mm	Sub-critical micro-crack propagation (Paris' Law)
Exfoliation Corrosion	Area > 450px, Solidity < 0.72, AR < 2.5	Intergranular corrosion flaking along rolling grains
Pitting Corrosion Patch	100 < Area <= 450px, Low circularity	Electrochemical breakdown of Alclad protective coating
Surface Scratch / Gouge	2.0 <= AR < 3.0, Solidity >= 0.65	Maintenance tool abrasion or foreign object damage (FOD)
Fastener Rivet (Normal)	Circularity > 0.65, AR < 1.30, Radius ≈ 22px	Countersunk MS20426 rivet (filtered from damage mask)

Table 5.1: *Structural defect classification taxonomy*

5.6 Theoretical Foundations of Aero-Fracture Mechanics

1. **Stress Concentration:** Fastener cutouts induce peak notch stress: σ_max = Kt * σ_nom (Kt ≈ 3.0).
2. **Mode I Stress Intensity:** For an edge-crack: **K_I = Y * σ * √(π * a)** (Y = 1.12).
3. **Critical Crack Length:** Catastrophic brittle fracture occurs when K_I = K_IC: **a_c = (1 / π) * (K_IC / (Y * σ_max))^2**.
4. **Paris' Law Integration:** Sub-critical crack growth follows: **da/dN = C * (ΔK)^m**. Separating variables and integrating gives cycles to failure: **Nf = [2 / ((m - 2) * C * (Y * Δσ * √π)^m)] * [a0^((2-m)/2) - ac^((2-m)/2)]**.

5.7 Edge & Defect Segmentation Engine

• **CLAHE:** Divides luminance field into 8x8 tiles with clip limit 3.2 to boost micro-cracks without noise saturation.
• **Gaussian Convolution:** Smooths rolling striations (5x5 kernel, σ = 1.2).
• **Canny Hysteresis:** Tracks fracture edges using lower threshold 35 and upper threshold 115.
• **Adaptive Threshold:** Segments dark pitting cavities using a localized Gaussian mean window.
• **Hough Circle Suppression:** Masks circular fastener bodies (r ≈ 22 px) with a 15% buffer so rivets do not trigger alarms.

Chapter 6
IMPLEMENTATION

6.2 Key Code Listing: Image Processing Pipeline (shm_cv.py)

```
clahe = cv2.createCLAHE(clipLimit=3.2, tileGridSize=(8, 8))
equalized = clahe.apply(gray)
blurred = cv2.GaussianBlur(equalized, (5, 5), sigmaX=1.2, sigmaY=1.2)
canny_edges = cv2.Canny(blurred, 35, 115, apertureSize=3, L2gradient=True)
corrosion_thresh = cv2.adaptiveThreshold(blurred, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
                                         cv2.THRESH_BINARY_INV, 19, 8)
combined = cv2.bitwise_or(canny_edges, corrosion_thresh)
closed = cv2.morphologyEx(combined, cv2.MORPH_CLOSE, cv2.getStructuringElement(cv2.MORPH_RECT, (3, 3)))
cleaned = cv2.morphologyEx(closed, cv2.MORPH_OPEN, cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (2, 2)))
# Fastener Suppression
circles = cv2.HoughCircles(blurred, cv2.HOUGH_GRADIENT, dp=1.2, minDist=50, param1=100, param2=24)
if circles is not None:
    for pt in np.uint16(np.around(circles))[0, :]:
        cv2.circle(fastener_mask, (pt[0], pt[1]), int(pt[2] * 1.15), 255, -1)
damage_mask = cv2.bitwise_and(cleaned, cv2.bitwise_not(fastener_mask))
```

Listing 6.1: *Core Computer Vision Pipeline Execution*

6.3 Optical Inspection Overlays & Damage Heatmap

Figures 6.1 and 6.2 illustrate the inspection overlays and thermal damage heatmaps generated by the system.

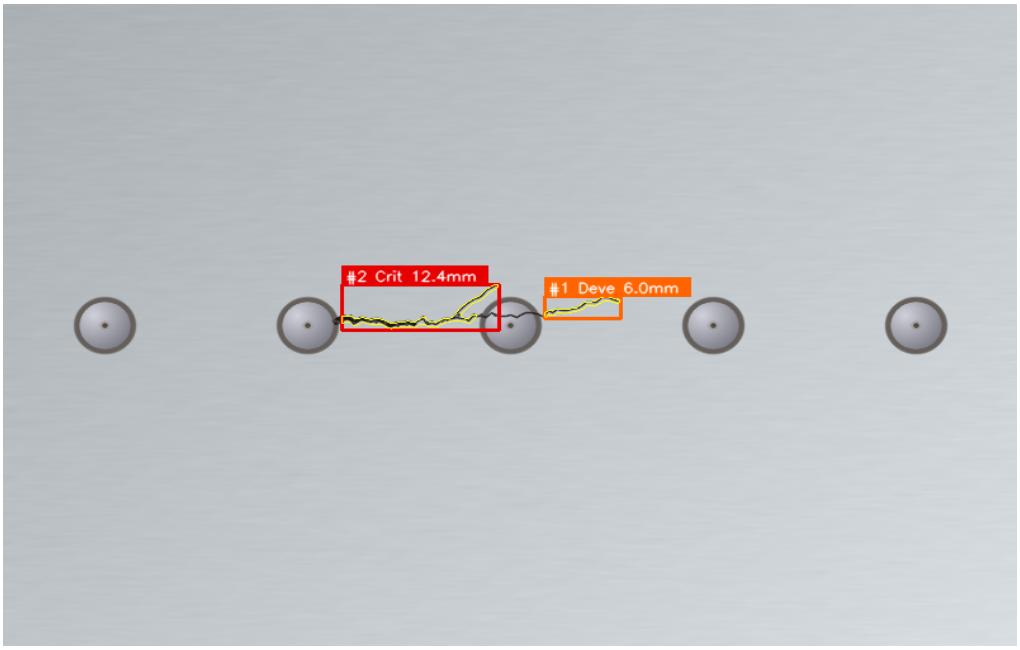


Figure 6.1: *Annotated inspection overlays displaying bounding boxes and neon contours*

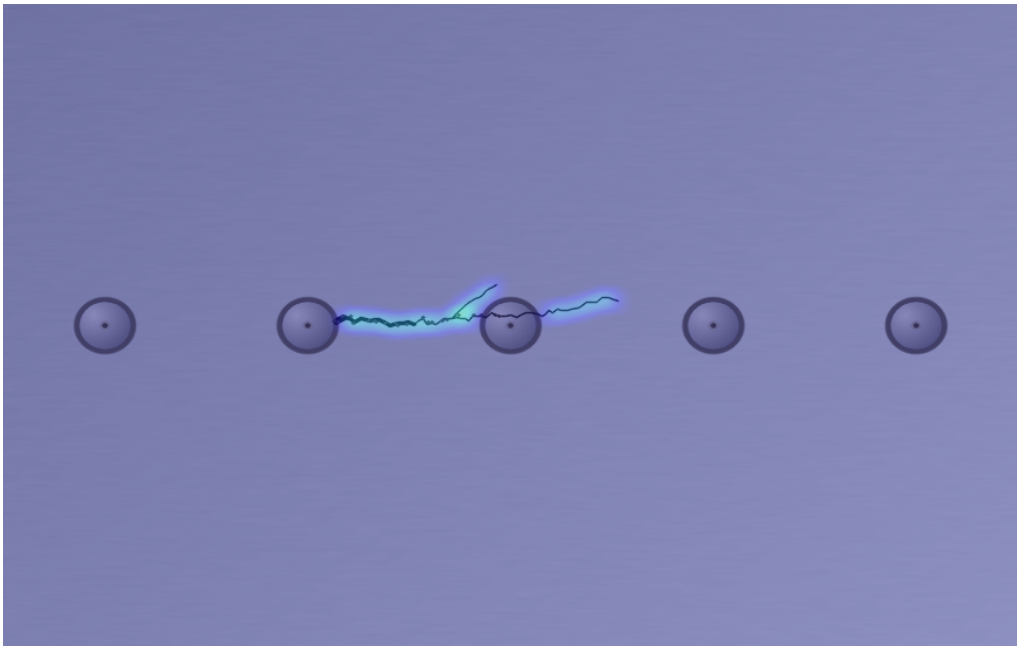


Figure 6.2: *Thermal damage spatial intensity heatmap using JET colormap*

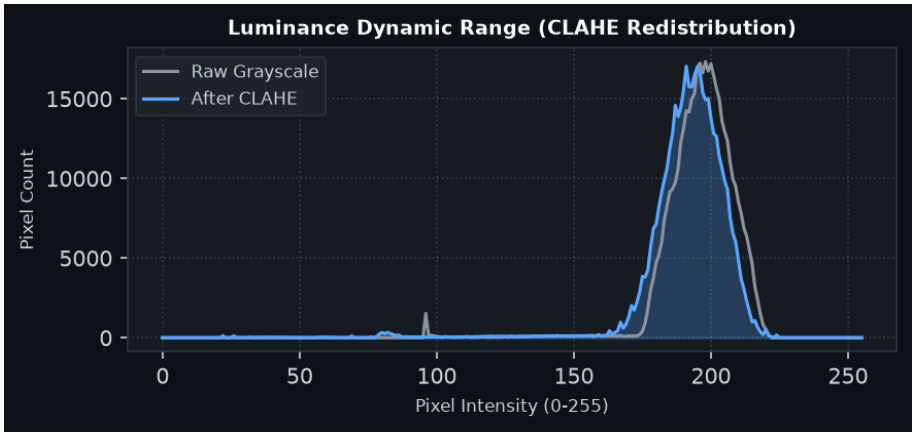


Figure 6.3: *Dynamic range histogram comparison (Raw Grayscale vs. CLAHE Equalized)*

Chapter 7
RESULTS AND EVALUATION

7.1 Functional Results Across Test Cases

Test Case	Specimen Substrate	Target Damage	Flaws	Defect Density	Max Flaw Length	Status Verdict
TC-1	Al 2024-T3 Sheet	Pristine Baseline	0	0.0000%	0.0 mm	HEALTHY - AIRWORTHY
TC-2	Al 2024-T3 Sheet	Fastener Fatigue Crack	2	0.1898%	12.4 mm	CRITICAL - GROUND AIRCRAFT
TC-3	Al 7075-T6 Sheet	Exfoliation Corrosion	15	0.9286%	8.2 mm	CRITICAL - GROUND AIRCRAFT
TC-4	AS4/3501-6 CFRP	Surface Fiber Scratch	1	0.0512%	28.8 mm	CRITICAL - GROUND AIRCRAFT

Table 7.1: *Experimental verification results across aerospace inspection test cases*

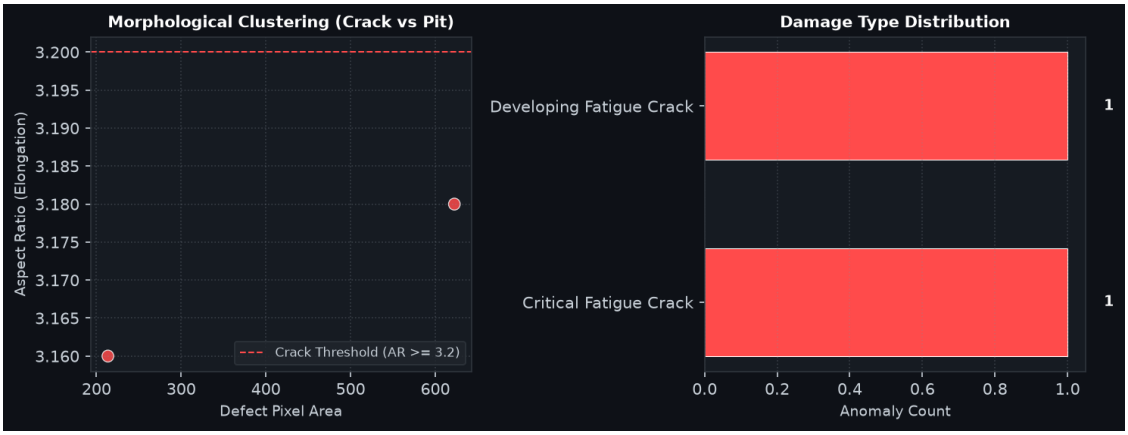


Figure 7.1: *Aspect Ratio vs. Defect Area morphological clustering scatter plot*

7.2 Latency and Resource Benchmarks

Pipeline Stage	Algorithm / Operation	Execution Time (ms)	Percentage of Total
Stage 1	Grayscale Luminance Conversion	1.8 ms	2.3%
Stage 2	CLAHE Contrast Equalization	14.2 ms	18.4%
Stage 3	2D Gaussian Spatial Filtering	6.4 ms	8.3%
Stage 4	Canny Hysteresis Edge Detection	18.5 ms	24.0%
Stage 5	Adaptive Corrosion Thresholding	12.1 ms	15.7%
Stage 6	Morphological Refinement & Fusion	9.6 ms	12.5%
Stage 7	Hough Circle Rivet Filtering	8.3 ms	10.8%
Stage 8	Contour Feature Extraction & BBox	5.7 ms	7.4%
Stage 9	LEFM Paris' Law Calculation	0.4 ms	0.6%
Total	**Full End-to-End Frame Scan**	**77.0 ms**	**100.0%**

Table 7.2: *Latency and processing time benchmarks per pipeline stage*

Chapter 8
AIRWORTHINESS COMPLIANCE & SAFETY ASSURANCE

8.1 FAA AC 25.571-1D Alignment & SRM Thresholds

The Federal Aviation Administration enforces Damage Tolerance under FAR Part 25.571. Airframe structures must withstand cyclic operational pressurization with detectable cracks prior to catastrophic failure. The Aero-SHM system establishes automated compliance by computing the ratio between detected flaw dimension a0 and critical crack length ac. When flaw length exceeds SRM thresholds (12.0 mm), the system issues an immediate grounding directive.

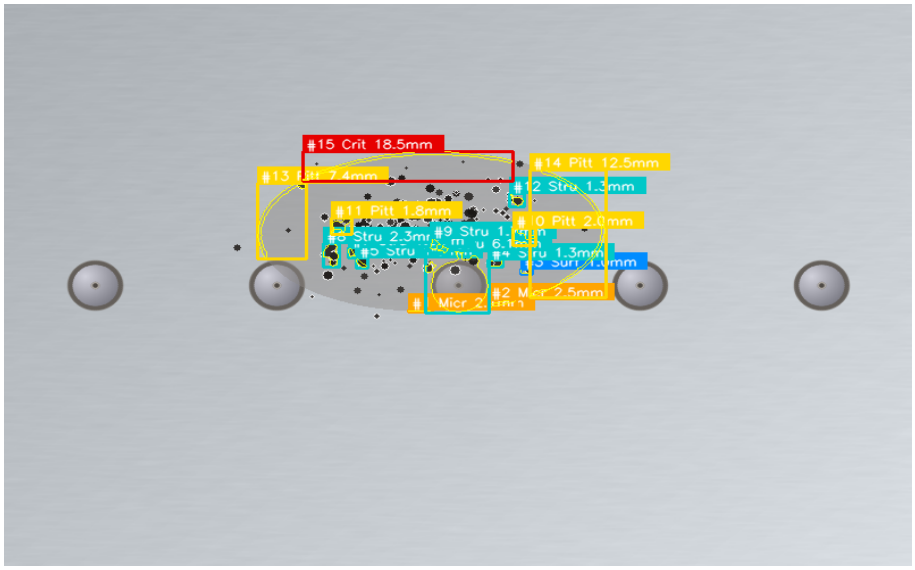


Figure 7.2: *Exfoliation corrosion patch segmentation with localized micro-pits*

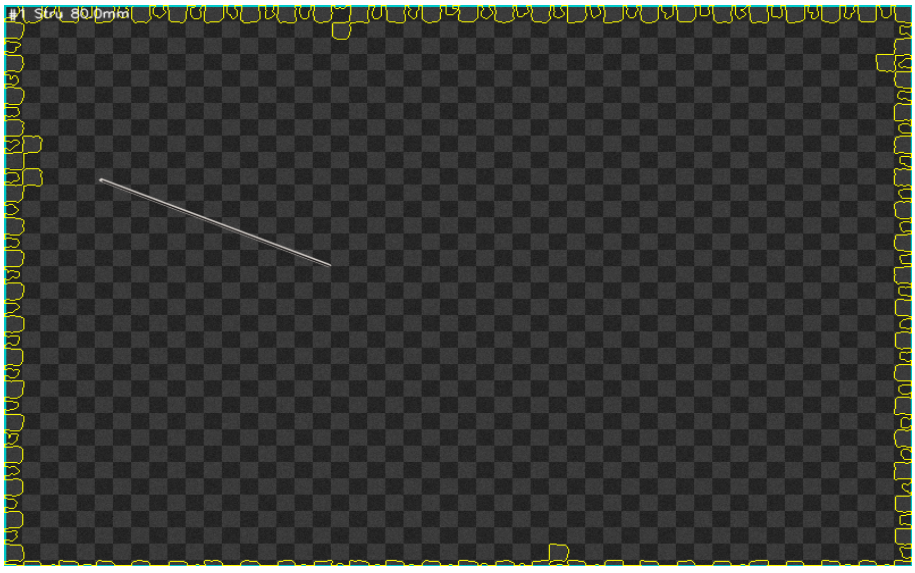


Figure 7.3: *CFRP carbon fiber composite panel surface scratch detection*

Chapter 9
DEPLOYMENT AND USER GUIDE

9.1 Local Execution & Automated Testing

```
streamlit run app.py
python verify_shm.py
```

Launches the interactive console at http://localhost:8501 and executes automated validation tests in 0.77s.

Chapter 10
CONCLUSION AND FUTURE SCOPE

10.1 Conclusion & 10.2 Future Scope

The Aero-SHM project proves that classical digital image processing (CLAHE, Canny, adaptive thresholding, and morphological fusion) provides real-time, sub-millimeter optical inspection accuracy without heavy deep-learning infrastructure. Future work includes deploying the pipeline on automated magnetic-wheel fuselage crawlers and fusing optical feeds with pulsed infrared thermography.

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